Cross-Sections of Commitment

Per-layer separation maps of attention morphology

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Abstract

Internal-state probes for hallucination detection are almost always read at a fixed depth — a layer chosen once, then held constant across models. That choice is rarely measured, and the cost of getting it wrong is unknown. We ask the question a deployer would ask: *if I must pick one layer, does any rule tell me which?* We answer it with registered per-layer sign-free AUROC curves of a sealed inter-head-disagreement metric, computed over every decoder block of 10 instruction-tuned models spanning four families and 28–88 blocks, on two hallucination-adjacent benchmarks. The confirmation bars were specified after the discovery grid was seen and frozen before any confirmation curve was inspected. Discovery on 8 cells and prospective held-out-model confirmation on 12 more give three results. First, **no placement rule was established**: peak location is model- and task-dependent, and neither the frozen absolute nor the frozen relative rule fired. Second, a sparse depth panel can **invert** a locus conclusion — a three-rung panel classified `Llama-3.3-70B` as a readout-locus model, while its full curve carries a broad mid-stack band peaking at 0.8975 on block 48/80, above the 0.8156 in-sample winner of the 29-cell panel that the rungs produced. Third, a terminal-block give-back appears in every evaluable held-out cell but one ($p_{\text{grid}} = 0.0005$, pooled dip CI $[0.123, 0.201]$) yet scores 8/12 against a frozen bar of $\geq 10/12$ plus family guards, and is therefore reported as **WEAKEN** — a strong, nearly-universal regularity whose registered confirmation bar it did not clear. The secondary cliff endpoint was gatekept behind that confirmation and is **NOT TESTED**. Three cells produced no number at all, for two distinct reasons — an operational gate, and twice an instrument-domain boundary where the sealed metric is undefined under total attention-sink collapse — and, separately, one fully evaluable cell is a genuine miss. For deployment the corollary is narrow and practical: measure the layer on the target model, because the registered tests established no rule that would let you transfer one.

1 Introduction

Probes are read at a depth, and the depth is usually guessed. Work that detects hallucination, deception, or failure from a model's internal state must choose where in the stack to look. In practice that choice is made once — a mid-stack layer, the last layer, a small sweep on one model — and then carried to other models as though it were a property of transformers rather than a property of the one model it was tuned on. The literature is candid that placement matters (Goldowsky-Dill et al., 2025; Ruan et al., 2026), but the shape of the underlying curve is rarely published, so the size of the risk is unmeasured.

The gap. A deployer choosing a probe layer has one question: is there a rule? Concretely — does separation live at a fixed absolute offset from the end of the stack, or at a fixed fraction of depth, or nowhere transferable at all? Answering it requires the whole curve, not a sample of it, and it requires deciding what would count as an answer *before* looking.

What we built. We compute a per-layer separation map: for every decoder block of a model, the sign-free AUROC of a sealed commit-time attention disagreement statistic against the hallucination label, plus a shuffled-label envelope that says what that block's AUROC looks like when the labels carry no information. The instrument is cheap — one forward pass per row, no training, no gradient — and it is registered. Grid A (8 cells) is *discovery*; grid B (12 cells from six held-out models in three registered families) is *prospective confirmation on two fixed benchmarks*, with every threshold, guard, and denominator frozen in a pre-registration commit before extraction began, and a scorer that runs exactly once.

Contributions.

1. **A blind-spot result** (Section 3): a three-rung depth panel inverted a locus conclusion on a 70B model. This bounds the interpretation of any fixed-layer probing claim, including our own prior panels, without changing their numbers.
2. **A registered placement result** (Section 5): no transferable depth rule was established at this resolution. We state this as *no rule established*, not as a proven non-law.
3. **A registered regularity, reported as a miss** (Section 4): the terminal-block give-back is nearly universal and misses its own confirmation bar. We report the bar and the miss together.
4. **Three failure modes, each a finding** (Section 6), including an instrument-domain boundary that every future use of the sealed metric must carry.

Scope, stated once and load-bearing. Every cell in this paper is a `torch`/Modal extraction. Eighteen of the twenty were specified `nf4`; both task cells of `Mistral-Medium-3.5` use FP8-origin weights deterministically dequantised for BF16 compute. None is byte-comparable with our sealed MLX panels, and the two are never pooled. Grid A rates are cited as discovery only and never enter a grid B count. Confirmation is claimed on two fixed benchmarks — adversarial NLI round 1 (Nie et al., 2020) and the QA split of HaluEval (Li et al., 2023) — and nothing here is task-general. We make no causal, lineage, or post-training claims, and no factorial family×scale claim: the comparative panel is unbalanced by construction.

2 The instrument

One forward pass per row, one number per block, and a null that says what noise looks like. For each input row we run the model to its commitment token and read, at every decoder block $\ell \in \{0, \ldots, N-1\}$, a sealed attention-morphology statistic: the Jensen–Shannon disagreement among that block's attention heads over the prompt, computed with the BOS position excluded (`final_js_no_bos`, our pre-registered primary). Three secondary metrics are banked alongside it and are not analysed here. "Block" and "decoder layer" denote the same object throughout.

Sign-free AUROC. Each cell contributes 200 rows, exactly 100 hallucinated and 100 faithful. For every block we compute $A_\ell = \max\{\mathrm{AUROC}_\ell, 1 - \mathrm{AUROC}_\ell\}$, which measures separation without committing to a direction. This is deliberate and it is a limitation we carry openly: sign-free AUROC is computed in-sample and cannot be deployed as-is, because the sign must itself be fit. It is the right instrument for the question *where is there signal at all*, and the wrong one for *how well would a probe here perform.*

The envelope. For each *grid-A* cell we redraw the labels 200 times under a fixed seed and record the 97.5th percentile of the resulting per-block sign-free AUROC. No envelope was registered for grid-B cells, and none is shown for them anywhere in this paper. This envelope is the paper's honesty floor: because A_ℓ is a maximum over two directions it is bounded below by 0.5 and drifts upward with finite samples, so a raw AUROC of 0.58 can be pure noise. Only blocks above their own envelope are treated as carrying signal. A *qualifying peak* ℓ^* is the block maximising A_ℓ subject to $A_\ell \geq 0.65$ and $A_\ell > \mathrm{env}_\ell$; a cell with no such block is undefined, and undefined counts as a failure in every registered denominator.

Two grids, and what separates them. Grid A (4 models × 2 tasks) is **discovery**: it is where the dip and cliff regularities were first seen, and its rates are cited as discovery throughout. Grid B (6 held-out models × 2 tasks = 12 cells) is **prospective held-out-model confirmation on two fixed benchmarks**. Grid B endpoints are measured with *cross-fitted* estimators: rows are split into $K = 5$ folds, the peak is located on each training fold, and the endpoint is evaluated on the held-out rows, which removes the selection bias that an in-sample argmax carries. Before grid B ran we re-scored grid A under exactly these estimators, so the discovery rates quoted for calibration (8/8 for the dip, 7/8 for the cliff) are measured with the same instrument as the confirmation.

What was frozen, and when. The grid-B pre-registration commit pins the endpoint definitions, the decision bars, the family guards, the denominator, the model revisions, and the extraction and scoring code together. Smoke runs are gates-only and never emit a per-block AUROC. The scorer runs *once*, after all twelve cells reach a terminal status. No grid-B threshold moved after the freeze and no grid-B endpoint was scored twice. The grid-A re-score described above is a distinct, pre-freeze calibration step and is always labelled as one.

3 The blind spot

A three-rung depth panel classified a model's locus, and the full curve says the classification was an artifact of where the rungs fell. Our own prior panels — and, we suspect, a good deal of fixed-layer probing work — sample depth at three points: the middle block, the penultimate block, and the last block ($\ell \in \{\lfloor N/2 \rfloor, N-2, N-1\}$). On `Llama-3.3-70B-Instruct/anli_r1` that panel found its best attention cell weak and its best readout cell strong, and the family was recorded as "readout locus".

Figure 1 shows what those three rungs were sampling. The full curve carries a broad mid-stack band: 44 of 80 blocks sit above the shuffled envelope, whose ceiling is 0.6145, and the qualifying peak is $A_{48} = 0.8975$. The three rungs land at blocks 40, 78 and 79. Two of them sit inside the envelope — statistically indistinguishable from shuffled labels — and the third clears it by about 0.02. The band was not merely under-sampled; it was missed on all three draws.

The comparison, stated carefully. The panel's winner was a *readout* cell scoring 0.8156 across a 29-cell panel, and $0.8975 > 0.8156$. Three qualifications travel with that inequality and none of them may be dropped. (i) Both numbers are *in-sample*: 0.8156 is the panel's in-sample full-panel marginal, quoted here for like-for-like comparison against an in-sample depth curve. The same winner's deployed out-of-bag median is 0.7954 with interval $[0.6995, 0.8728]$ and winner stability 0.511, so a reader must not take 0.8975 as beating a bootstrapped bound. (ii) The two quantities come from *different instruments*: a readout cell read at the commitment step is not an attention-morphology cell read at a block, and Fig. 1 is therefore a cross-instrument comparison. (iii) 0.8156 is the winner of the full 29-cell panel, not of its readout subset, so the accurate phrasing is that the mid-stack peak exceeds *the panel winner*.

What this does and does not license. It does not change a single panel number, and it does not falsify the panel's winner: on the rungs it sampled, the panel reported correctly. What it bounds is *interpretation*. "This family reads out late" was never measured; it was inferred from three samples that happened to straddle a band. Any claim of the form "family F localises signal at region R", ours included, inherits this caveat unless the curve behind it was measured. The same model's `halueval_qa` curve carries a band that is broader and earlier still (54/80 blocks above envelope, $\ell^* = 26$), so this is not a single-cell accident.

4 The terminal give-back: a regularity that missed its bar

The primary endpoint scores $8/12$ against a frozen bar of $\geq 10/12$ plus family guards, and is therefore WEAKEN, not CONFIRM. We state the miss before the anatomy, because the anatomy is unusually favourable and would otherwise read as a rescue.

What was registered. E5 asks whether separation gives back at the very end of the stack: whether the final block scores materially below the qualifying peak. A cell succeeds when the cross-fitted dip magnitude

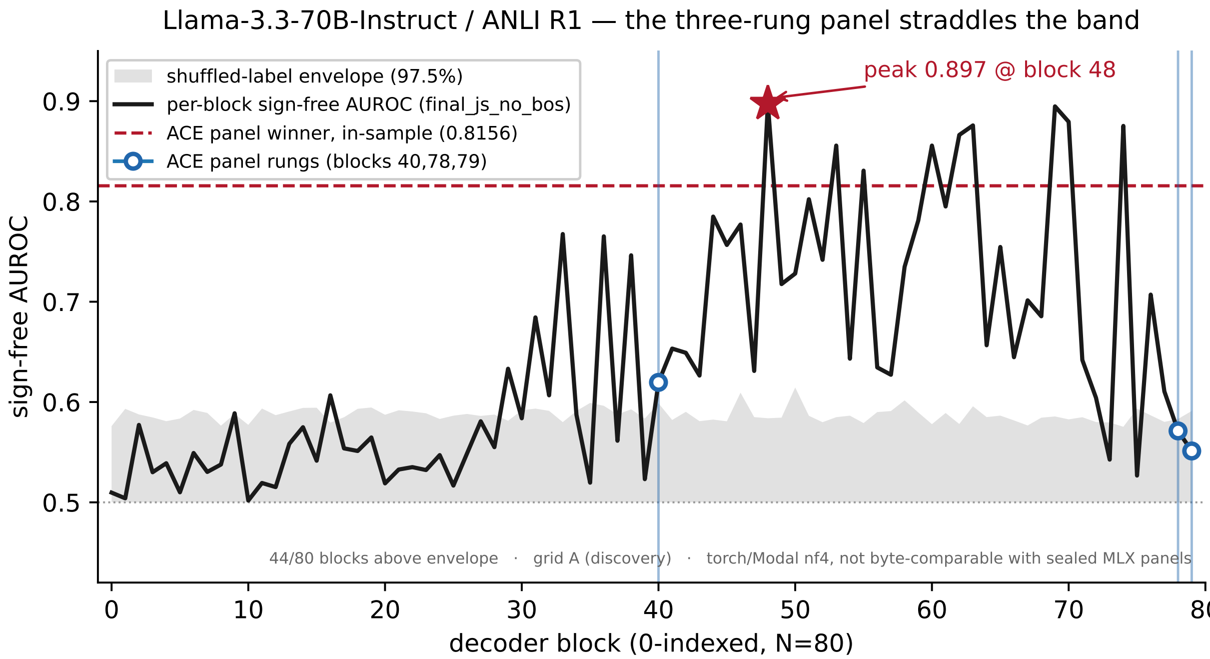

Figure 1: The blind spot. Per-block sign-free AUROC for `Llama-3.3-70B-Instruct/anli_r1` (grid A, discovery). Shaded band is the 97.5% shuffled-label envelope. Circles mark the three rungs a fixed-depth panel samples (40, 78, 79 of 80); two lie inside the envelope. The dashed line is the in-sample winner of that panel's full 29 cells (0.8156, a readout cell read at the commitment step — a different instrument, shown for scale, not as a matched baseline). Both numbers are in-sample: the same winner's deployed out-of-bag median is 0.7954 $[0.6995, 0.8728]$ with stability 0.511, so the peak does not beat a bootstrapped bound. The qualifying peak is 0.8975 at block 48. Grid A only; `torch`/Modal `nf4`, not byte-comparable with sealed MLX panels.

$\Delta_{cf} \geq 0.05$ with all five folds qualifying. The decision rule, frozen before extraction: **CONFIRM** at $\geq 10/12$ successes *and* $\geq 3/4$ in each of the three grid-B families *and* pooled $p_{\text{grid}} < 0.05$; **WEAKEN** at 8–9/12, or $\geq$ 10/12 with a family or p failure; **FALSIFY** at $\leq 7/12$. The denominator is always 12: missing, gate-aborted, and no-qualifying-peak cells count as failures.

What happened. Eight of twelve. Families 4/2/2 against a $\geq 3/4$ guard that two of three families miss. Pooled $p_{\text{grid}} = 0.0005$. Two of the three CONFIRM conditions therefore fail, and the registered decision is **WEAKEN**.

The anatomy, labelled descriptive. Of the nine evaluable cells, eight show the dip; the pooled dip interval is $[0.123, 0.201]$; the registered evaluable-only sensitivity recount is 8/9 and the leave-Medium-out recount is 6/10. Figure 2 shows every cell with its interval. This is the honest summary: *the terminal give-back appears in every evaluable held-out cell but one, and the registered bar priced in exactly the failures that materialised.* The frozen denominator counts the three aborted cells against the endpoint by design, precisely so that an instrument that fails to produce a number cannot be quietly excluded from its own test. We report it as written — a strong, nearly-universal regularity whose registered confirmation bar it did not clear. Neither the p-value nor the 8/9 sensitivity is a softer bar, and neither is offered as one.

The one true miss. `Mistral-Small-3.2-24B/halueval_qa` returns $\Delta_{cf} = 0.0045$ — not a failure of measurement but a cell where the terminal give-back genuinely is not there. It sits essentially on zero in Fig. 2 and it is the reason the regularity is "nearly" universal rather than universal.

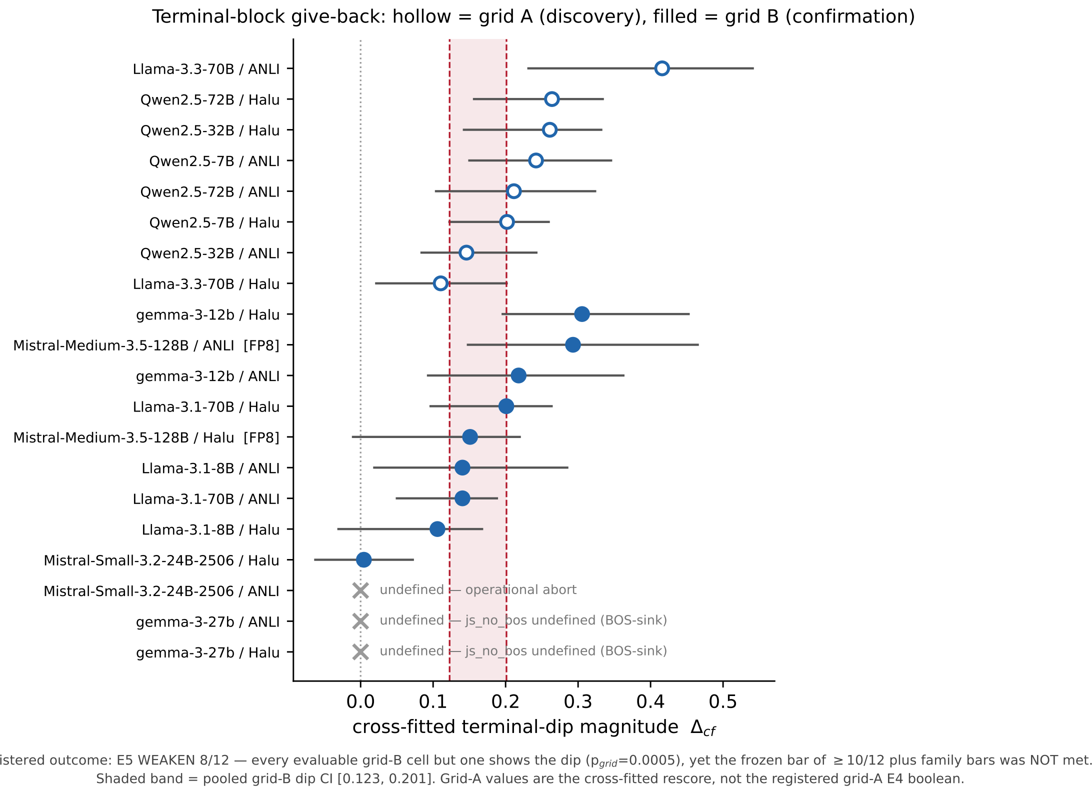

Figure 2: Terminal-block give-back, all cells. Cross-fitted dip magnitude Δ_{cf} with 5–95 intervals. Hollow markers are grid A (discovery, re-scored under the grid-B estimators — *not* the registered grid-A boolean); filled markers are grid B (confirmation). Shaded band is the pooled grid-B interval $[0.123, 0.201]$. The three undefined cells are shown with their reasons distinguished and are counted as failures in the 12-cell denominator. `Mistral-Medium-3.5` is FP8-origin.

The cliff endpoint was not tested

E6 is NOT TESTED, and no cliff result is reported in this paper. The registration gatekeeps the cliff endpoint behind an E5 CONFIRM. E5 did not CONFIRM, so the gate closed and no confirmatory cliff evaluation ever ran. This is not a null result and it must not be read as one: a gatekept endpoint that was never evaluated has no verdict at all.

What we may show is the grid-A discovery observation that motivated registering it. Figure 3 plots, per grid-A cell, the largest single-block jump J against the total rise R. In seven of eight discovery cells the rise arrives in roughly one block. That is *discovery only*; it carries no confirmatory status, and the grid-B descriptive rate (2/12, depressed by early-peak cells that the registration defines as cliff-undefined) is reported without interpretation. One grid-A cell is excluded from Fig. 3 entirely: `Llama-3.3-70B/halueval_qa` reports $J = 0.0000$, but that zero is a structural empty-window fallback — its peak at block 26 falls below the jump-search floor, leaving the search segment empty — and plotting it as a measured absence of a jump would be a fabrication.

5 Placement: no rule established

Neither frozen placement rule fired, and the descriptive panel beside them tests no law at all. Grid A registered two candidate rules and a tie-breaker between them. An *absolute* rule — separation peaks a fixed number of blocks from the end — required the spread of $N - \ell^*$ to be at most 2 blocks. A *relative*

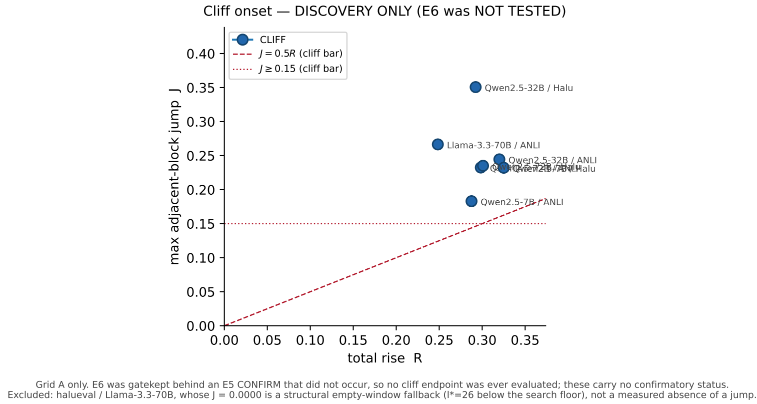

Figure 3: Cliff onset — DISCOVERY ONLY. Grid A cells: largest adjacent-block jump J against total rise R, with the two registered cliff bars drawn. E6 was gatekept behind an E5 CONFIRM that did not occur, so no cliff endpoint was ever evaluated and these points carry no confirmatory status. `Llama-3.3-70B/halueval_qa` is excluded: its $J = 0.0000$ is a structural empty-window fallback, not a measured zero.

rule — separation peaks at a fixed fraction of depth — required the spread of ℓ^*/N to be at most 0.015. Neither came close, and the registered outcome is **UNDECIDED**: at this resolution, *no placement rule was established.* Note the scope precisely — E1 is a *grid-A* endpoint, so the frozen rules were tested on four discovery models, not on all ten. The ten-model panel below is registered as descriptive and is explicitly forbidden from making law or non-law claims. We phrase it that way deliberately. Failing to establish a rule — on four discovery models at $n = 200$ per cell, with a ten-model descriptive panel that offers no obvious replacement — is not the same as proving no rule exists, and this paper does not claim the stronger thing.

An artifact that looked like a law. Before the full curves existed, a three-rung panel appeared to show separation peaking at $N-2$. It does not, and the reason is the sampling failure Section 3 documents in its other guise: $N-2$ merely won among three sampled points. On the full curves the six `Qwen2.5` peaks are spread from $N-2$ to $N-17$, at depth fractions 0.79 to 0.93 — one of them does land at $N-2$, but a spread of 15 blocks is far too wide for the frozen absolute rule, and a fraction range that broad is far too wide for the relative one.

The panel. Figure 4 plots cross-fitted peak fraction against depth for all 17 evaluable cells. Grid A and grid B are drawn separately and are never pooled numerically. There are no confidence intervals on these points, and their absence is deliberate: no bootstrap interval for this estimator is banked on either grid, and grid A's peak-location interval belongs to a *different* estimator (the in-sample argmax), so splicing it onto cross-fitted points would silently mix two things. We plot instead the spread of the five cross-fitted training-fold peaks, which is what we actually have.

Family structure, descriptive only. Mean peak fractions cluster by family: `Mistral` 0.88 and `Qwen` 0.85 peak late, `Llama-3.1` 0.41 and `Llama-3.3` 0.46 peak mid, `Gemma` 0.64 falls between. This is structure, and it is not a rule — with two models per family, family is fully confounded with tokenizer, architecture, and size, and we make no factorial claim. Two registered descriptive predictions about the Llama band were both borne out *on `anli_r1`*, the only task over which they were registered: the qualifying-block count in the mid region was ≥ 10 in a majority of folds for `Llama-3.1-70B`, and < 5 in a majority for `Llama-3.1-8B`. The registration forbids "replicates", "stable" and "emergent" vocabulary here and we honour that: this is a same-size version comparison in which base-checkpoint identity was never established.

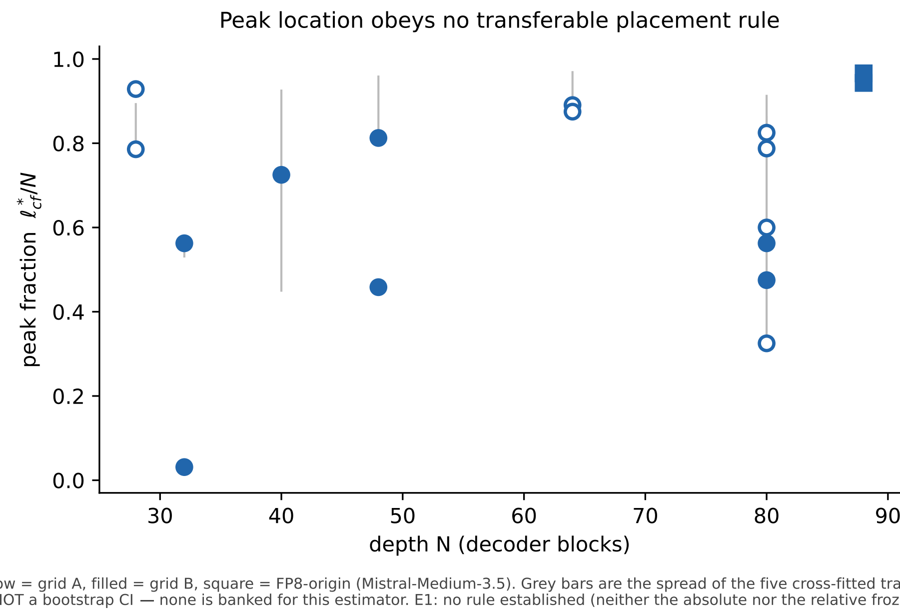

Figure 4: No placement rule established. Cross-fitted peak fraction ℓ^*_{cf}/N against depth N. Hollow = grid A, filled = grid B, square = FP8-origin. Grey bars are the spread of the five cross-fitted training-fold peaks, *not* a bootstrap interval — none is banked for this estimator. Grid A and grid B are shown together for visual context only and are never pooled numerically.

6 Three failure modes, three findings

Three of twelve grid-B cells produced no number, and the three reasons are scientifically distinct. They count identically against the endpoint — all three are failures in the frozen denominator — but conflating them in the discussion would hide the most transferable result in this paper.

A behavioral gate (operational). `Mistral-Small-3.2-24B/anli_r1` never produced a curve: the extraction aborted on a missing prompt manifest. This is an operational failure with no bearing on the instrument, and we record it as one. It counts as a failure because the registration says every non-producing cell does.

An instrument-domain boundary (the transferable one). Both `gemma-3-27b-it` cells aborted with the sealed primary metric returning *undefined* at block 3, row 0. The cause is not a bug: under a total attention-sink collapse, essentially all attention mass concentrates on the BOS position, and a statistic defined on the non-BOS distribution has no distribution left to measure. `final_js_no_bos` therefore has a *domain*, and `gemma-3-27b` sits outside it.

This is a standing caveat for every future use of the metric, in this paper and elsewhere: where the sink collapses, the value must be rendered and reported as explicitly undefined — never as zero, and never silently dropped. A zero would read as "the heads agree completely", which is the opposite of what the model is doing; a dropped cell would quietly shrink a denominator. Both figures and tables in this paper mark these cells undefined for exactly this reason.

One true miss (the honest one). The third is not a failure of the apparatus at all: `Mistral-Small-3.2-24B/halueval_qa` produced a clean curve whose terminal dip is $\Delta_{cf} = 0.0045$. The regularity is simply absent there. A registered endpoint that never encounters a genuine miss has not been tested very hard.

cell	grid	N	peak blk	peak CI	peak AUROC	mid med	E2	Δ_{cf}
Llama-3.3-70B / ANLI	A	80	48	[48, 74]	0.8975	0.649	CLIFF	0.4160
Qwen2.5-32B / ANLI	A	64	56	[56, 62]	0.8655	0.545	CLIFF	0.1460
Qwen2.5-72B / ANLI	A	80	66	[66, 77]	0.8376	0.539	CLIFF	0.2115
Qwen2.5-7B / ANLI	A	28	22	[22, 25]	0.8336	0.546	CLIFF	0.2420
Llama-3.3-70B / Halu	A	80	26	[26, 48]	0.8621	0.721	GRADUAL	0.1105
Qwen2.5-32B / Halu	A	64	56	[35, 61]	0.9106	0.618	CLIFF	0.2610
Qwen2.5-72B / Halu	A	80	63	[63, 63]	0.8961	0.595	CLIFF	0.2640
Qwen2.5-7B / Halu	A	28	26	[26, 26]	0.8713	0.546	CLIFF	0.2020
Llama-3.1-70B / ANLI	B	80	45	–	–	–	–	0.1405
Llama-3.1-8B / ANLI	B	32	18	–	–	–	–	0.1405
Mistral-Medium-3.5-128B / ANLI	B	88	83	–	–	–	–	0.2930
Mistral-Small-3.2-24B-2506 / ANLI	B	40	–	–	–	–	–	undef.
gemma-3-12b / ANLI	B	48	39	–	–	–	–	0.2180
gemma-3-27b / ANLI	B	62	–	–	–	–	–	undef.
Llama-3.1-70B / Halu	B	80	38	–	–	–	–	0.2010
Llama-3.1-8B / Halu	B	32	1	–	–	–	–	0.1060
Mistral-Medium-3.5-128B / Halu	B	88	85	–	–	–	–	0.1510
Mistral-Small-3.2-24B-2506 / Halu	B	40	29	–	–	–	–	0.0045
gemma-3-12b / Halu	B	48	22	–	–	–	–	0.3055
gemma-3-27b / Halu	B	62	–	–	–	–	–	undef.

Table 1: Per-cell verdict table. `-` means *not scored for that grid*, not missing data. Grid-A peak block is the in-sample ℓ^*; grid-B peak block is the cross-fitted ℓ^*_{cf} — a different estimator, never compared numerically across the two. Δ_{cf} for grid A is the cross-fitted re-score, not the registered grid-A boolean. `Mistral-Medium-3.5` is FP8-origin.

7 The full picture

Figure 5 shows every curve in the study. Table 1 gives the per-cell numbers and Table 2 gives every registered endpoint beside its frozen bar and its actual outcome. Two reading instructions travel with them. First, grid A panels carry their registered shuffled envelope and grid B panels do not: no envelope was ever registered for grid-B slugs, and computing one after the verdict would be a new Monte-Carlo statistic on confirmatory cells. Grid-B curves are recomputed from the banked extractions through the registered scorer's own validating loader — which re-checks the frozen data hash, the pinned model revision, and both extraction gates — but they are shown unshaded, and their absence of shading is a statement about the registration, not about the data. These recomputed curves are **post-verdict descriptive rendering and carry no confirmatory status**: the registered grid-B scorer banked no per-block curve, and nothing in Section 4 or Section 5 depends on them. Second, several columns of Table 1 are empty for grid B. They are marked `-`, meaning *not scored for that grid*, not *missing*: peak AUROC, mid-region median, rise-shape and peak-location interval were never computed for grid B, and backfilling any of them now would be new computation on confirmatory cells.

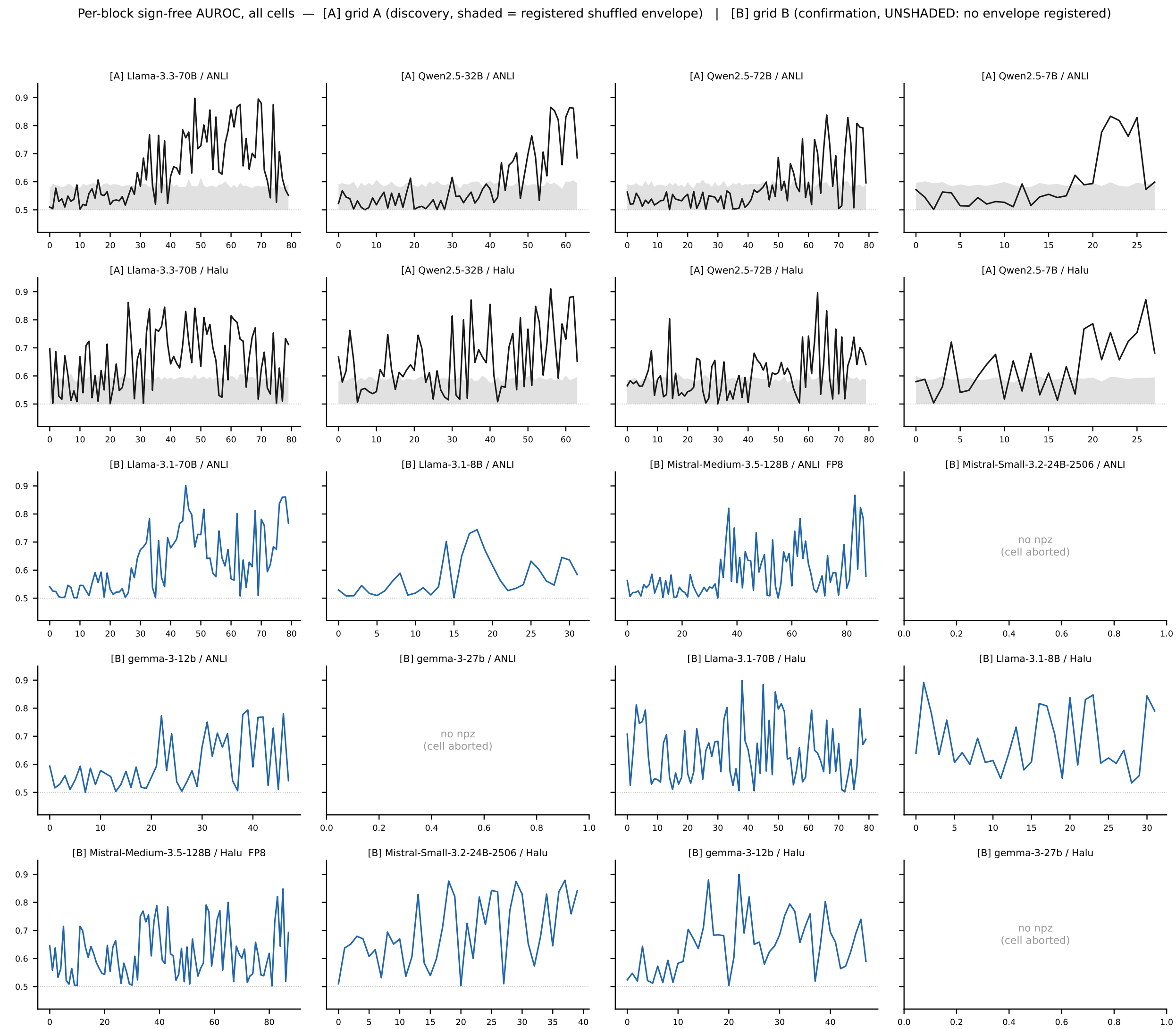

Figure 5: Every depth curve in the study. Per-block sign-free AUROC. Top block, black: grid A (discovery), shaded with its registered 97.5% shuffled-label envelope. Lower block, blue: the grid-B *cells*, drawn from curves recomputed after the single-look scoring — these curves are post-verdict descriptive rendering and carry no confirmatory status, and they are deliberately unshaded because no envelope was registered for grid-B cells. Three grid-B cells produced no extraction and are shown as labelled empty panels rather than omitted. The two grids are never pooled numerically.

id	endpoint	frozen bar	actual	outcome
E5	terminal-block give-back	CONFIRM if $\geq$10/12 AND $\geq$3/4 in each of 3 families AND p_{grid} < 0.05; WEAKEN if 8–9/12; FALSIFY if $\leq$7/12	8/12; families 4/2/2; p_{grid}=0.0005	WEAKEN – count and bars both missed
E6	cliff onset	gatekept on E5 == CONFIRM; then $\geq$9/12 AND $\geq$4/6 per task AND $\geq$2/4 per family AND p_{grid} < 0.05	gate closed (E5 did not CONFIRM); descriptive-only count 2/12 carries no verdict	NOT TESTED – no cliff was ever evaluated
E7	cross-task peak distance	descriptive-registered, no bar, no verdict vocabulary	6 models reported	descriptive
E8	Llama mid-stack band context	descriptive-registered, no bar	P8 hit on ANLI ($\geq$10 qualifying mid-blocks, majority of folds, 3.1-70B); P9 hit on ANLI (<5, 3.1-8B)	descriptive
E1”	peak-fraction panel	descriptive-registered, no bar	no transferable placement rule fired	descriptive – no rule esta

Table 2: Registered endpoint ledger. Every endpoint with its frozen bar and its actual outcome. The denominator is always 12, including the three aborted cells.

8 Related work: who measures the layer, and who assumes it

Four recent internal-state detectors span the full range from measuring the probe layer to assuming it, and the assumption is the more common practice. We group them on that axis because it is the axis this paper is about. All four study *deception* rather than hallucination, and three read the residual stream rather than attention morphology, so none is a replication of anything here; they are convergent evidence about *depth*.

Measured, and closest to what we plot. Dombrowski & Corlouer (2024) trace information-theoretic quantities — predicted-token probability, entropy, and KL divergence to the final layer — through *every* layer via the logit lens, on a Mistral-lineage 7B and two Llama-2 chat models, read at the token immediately *before* the model commits to a true or false completion. That is the object shape of Fig. 1: a per-layer curve of a scalar, compared across a binary condition, at the commitment position. Two findings bear on ours. On their stronger datasets separation opens mid-stack, around layers 15–20 of 32, not at the terminal block. And its strength is *dataset-dependent*: on `FreebaseStatements` the truth and lie conditions only separate at layer 28, which they attribute to the model being less certain of the truth in that context. A placement rule would have to survive that dependence, and Section 5 reports that ours did not. They also note that earlier probing work reads hidden states *after* the statement is produced — an independent statement of the commit-locus position measured here. Their instrument is the unembedded predictive distribution and ours is attention morphology, so the correspondence holds at the level of depth phenomenology only.

Measured, and deployed. Goldowsky-Dill et al. (2025) fit a deception probe in the `Llama-3.3-70B` residual stream at layer 22 of 80, chosen by sweep; their appendix shows the usable deployment band is mid-stack and narrow, with recall at a fixed low false-positive rate collapsing within a couple of layers while AUROC stays respectable. Ruan et al. (2026) make the sweep part of the method: probe layer is fixed *per model* by a preliminary per-layer AUC sweep, landing at 14/28 for a 3B Llama and 20/28 for a 7B Qwen. That is Section 9's corollary already in production.

Assumed. Hollinsworth et al. (2026) scale a lie-detector-gated oversight protocol across Llama-3 models to 405B parameters and Qwen-3 models to 32B, and report a genuine scaling trend: undetected deception falls from 34% to 14% at a fixed 99% detector true-positive rate. Their probe is read at $\lfloor L/2 \rfloor$ — a fixed depth fraction applied across two families spanning under a billion to 405 billion parameters. We find no layer ablation and no AUC-versus-layer curve reported in that paper. We flag this precisely, because it is the kind of choice this paper is about and not a claim we can refute: absent a sweep there is no number to check $\lfloor L/2 \rfloor$ against. What our results supply is a testable prediction. Our registered peak fractions cluster by family (Fig. 4), with the `Qwen` cells late in the stack — mean fraction 0.85 — and the Llama cells mid-stack. If detector accuracy were ever plotted against layer at fixed model size in that setting, a mid-stack read should cost more for Qwen-3 than for Llama-3. That is a prediction about a different construct on a different object, offered as such.

Across the group: where placement is measured it is reported as model-specific and the usable band is narrow; where it is assumed, it is assumed to be a constant fraction of depth. None of the four reports a sweep supporting that constant. Section 5 is the nearest test of the constant-fraction idea we are aware of — on a different construct and a different object, and neither frozen rule fired.

9 What this means for probe placement

Measure the layer; do not inherit it. The deployment corollary is narrow, and it is the practical point of the paper. If the registered absolute and relative rules were both left unestablished on the discovery grid, and the descriptive ten-model panel offers no obvious replacement, then a probe layer imported from another model — or from another size in the same family — is an untested assumption. The replacement is cheap: one per-layer sweep on the target model, on a few hundred labelled rows, costs a single forward pass per row and produces exactly Fig. 1 for that deployment. Section 8 shows this is already deployed practice wherever anyone has looked: Ruan et al. (2026) fix the probe layer per model by exactly such a sweep, and one of their two models, `Qwen2.5-7B`, is in our own panel, where the peak is likewise late.

The cost of not measuring is documented next door. Goldowsky-Dill et al. (2025) find the usable band mid-stack and narrow, with recall at a fixed low false-positive rate collapsing within a couple of layers while AUROC stays respectable — the same failure mode we report from the other side: an aggregate score can look healthy at a layer where the deployable operating point has already gone.

10 Limitations

The honest list, and it is long enough to matter.

- **Sign-free and in-sample.** A_ℓ maximises over both directions and is computed in-sample. It answers *where is there signal*, not *how well would a probe do.* No number here is a deployment estimate.
- $n = 200$ **per cell**, 100/100. Peak locations are noisy at this resolution; the failure to establish a placement rule is a statement at *this* resolution and a denser study could find one.
- **Two benchmarks.** Confirmation is claimed on `anli_r1` and `halueval_qa` and is not task-general.
- **Unbalanced panel.** Between one and three evaluable models per family, family confounded with tokenizer, architecture and size. No factorial claim; the version comparison establishes no base-checkpoint identity.
- **One precision-heterogeneous model, two cells.** `Mistral-Medium-3.5` contributes both of its task cells from FP8-origin weights, deterministically dequantised to BF16, and is flagged wherever it appears; the registered leave-Medium-out sensitivity is reported alongside the verdict.
- **Non-byte-comparable throughout.** All cells are `torch`/Modal and are never pooled with sealed MLX panels.
- **Per-cell argmax.** Peak selection is a maximum over N blocks. The envelope and the cross-fitting mitigate the resulting optimism; they do not eliminate it.
- **Non-random evaluability.** Only 9 of 12 grid-B cells produced a number, and the three that did not are not a random sample: two are the same model failing the same instrument boundary. The endpoint counts them as failures, which is conservative, but the surviving nine are not an unbiased draw.
- **E1's frozen test covered four models.** The absolute and relative placement rules were grid-A endpoints. The ten-model panel is descriptive and tests no rule.

- **Grid-B full curves are post-verdict.** They were recomputed for Fig. 5 after the single-look scoring and carry no confirmatory status; no endpoint depends on them.
- **No mechanism.** We report where separation lives, not why. Nothing here identifies a computation.
- **A frontier-scale cell was planned and never run.** A 405B extraction was registered as a descriptive stretch, sat outside every confirmatory denominator by construction, and was not executed. No number in this paper depends on it.

Reproducibility

Pre-registrations, extraction and scoring code, the banked per-cell arrays, and the figure builder — which asserts every plotted value against the scored artifacts before writing a file — are in the project repository at `https://github.com/flowstyleliving/commit-confluence`, under `exploratory/depth-curve/`.

References

N. Goldowsky-Dill, B. Chughtai, S. Heimersheim, and M. Hobbhahn. Detecting strategic deception using linear probes. *arXiv:2502.03407*, 2025.

J. Ruan, L. Huang, Y. Zhou, X. Wei, H. Wang, and M. Sun. Doomed from the start: Early abort of LLM agent episodes via a recall-controlled probe cascade. *arXiv:2607.06503*, 2026.

A.-K. Dombrowski and G. Corlouer. An information-theoretic study of lying in LLMs. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235, 2024.

O. J. Hollinsworth, A.-K. Dombrowski, S. Adam-Day, A. Gleave, and C. Cundy. Scaling trends for lie detector oversight in preference learning. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, PMLR 306, 2026. arXiv:2607.01567.

Y. Nie, A. Williams, E. Dinan, M. Bansal, J. Weston, and D. Kiela. Adversarial NLI: A new benchmark for natural language understanding. In *ACL*, 2020.

J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie, and J.-R. Wen. HaluEval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, 2023.